

# RYAN NG JINGKAI

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## EDUCATION

**National University of Singapore**

**Expected Jun 2028**

**Bachelor of Engineering in Computer Engineering + Master of Science in Computer Engineering, 3+1 Programme**

- Major in Computer Engineering (First Class Honours), Concentration in Large-Scale Computing and Communication & Networking
- Honors & Awards: Engineering Scholars Programme; Wee Hur Prize — ranked Top 3 among 626 students for PF1101, AY24/25 Semester 1
- Relevant coursework: Real-Time Operating Systems, Software Engineering & Object-Oriented Programming, Data Structures and Algorithm, Computer Architecture

## EXPERIENCE

**TikTok, Software Development Engineer in Test Intern**

**Jul 2026 - Present**

- Developed and debugged automated testing solutions; implemented platform reliability improvements across distributed service infrastructure; and established AI-assisted workflows to accelerate test-suite development and root-cause analysis.

**Synapse, DevOps Engineer Intern**

**Jan 2026 - Jul 2026**

- Automated system administration workflows across Windows and Linux environments using Ansible, reducing manual configuration and health-check time by up to 80% across ~3000 servers
- Designed and developed a centralized CMDB and internal operations tool for server management, focusing on backend API design and database schema design, standardizing infrastructure data across 200+ servers and improved visibility for ~20 engineers
- Implemented and optimised CI/CD pipelines to automate application testing, Docker image builds, image publishing, and deployment of containerized applications to an OpenShift cluster, reducing manual release steps and improving deployment consistency
- Built an internal VM-based CI tool for Ansible playbook testing by creating custom VirtualBox base images, automating clone provisioning and lifecycle management, playbook execution, and output capture for validation

**Schneider Electric, Software Developer Intern**

**Jul 2025 - Oct 2025**

- Developed and deployed embedded software on STM32 using Zephyr RTOS, integrating an HTTP server, file system, and the LLEXT subsystem for extensible runtime functionality
- Engineered a secure bootloader for STM32 using Zephyr RTOS, implementing RSA-2048 signature verification and SHA-256 hashing to ensure 100% firmware integrity during the boot process
- Created automation scripts to streamline the software build pipeline, reducing manual workload by 75% and ensuring consistency across all build processes

## PROJECTS

**Homelab**

**Jan 2026 - Present**

- Built and administered a multi-node self-hosted Kubernetes cluster using kubeadm, containerd and Calico, and provisioned and configured supporting cloud VPS infrastructure using Terraform and Ansible
- Configured Tailscale, Cloudflare Tunnel and Kubernetes ingress controllers to provide secure private access and public domain routing to hosted services
- Implemented Prometheus-based metrics collection and Grafana dashboards to monitor cluster nodes, workloads and application health
- Optimised GitHub Actions CI pipelines to automate application testing and container image publishing to GHCR, with Argo CD managing GitOps-based Kubernetes deployments

**MigrateOS (Python Migration Tool)**

**May 2025 - Present**

- Developed a Python automation tool to streamline EOSL server migration by generating Ansible playbooks that capture and reproduce legacy system configurations

**Operation Einstein (Student Progress Tracking Platform)**

**Aug 2025 - Present**

- Built a RESTful backend in Node.js and Express with 20+ endpoints for core application workflows, and designed a Supabase (PostgreSQL) database from scratch, optimizing API and query performance to support data integrity. Services can support up to 50 concurrent users with sub-200 ms latency
- Dockerized services with a CI/CD pipeline to automatically test and deploy updates to self-hosted infrastructure

## SKILLS

Languages: C, C++, Python, Java, Verilog, SQL, HTML, CSS, JavaScript, TypeScript

DevOps/Infrastructure: Ansible, Docker, OpenShift/Kubernetes, Terraform, Git, GitHub Actions (CI/CD), Argo CD, Linux, Prometheus, Grafana, Windows Server, VirtualBox

Frameworks and Tools: Node.js, Express, Flask, Supabase/PostgreSQL, React, Next.js, Supabase/PostgreSQL